DrugWiser: Machine Learning-Based Personalized Medicine Recommendation System

Robert G. de Luna
Polytechnic University of the Philippines
Sto. Tomas City, Batangas, Philippines
rgdeluna@pup.edu.ph

Ann Margaret J. Ambasa
Polytechnic University of the Philippines
Sto.Tomas City, Batangas, Philippines
annmargaretjambasa@iskolarngbayan.pup.edu.ph

Tyrone Paolo V. Garcia
Polytechnic University of the Philippines
Sto.Tomas City, Batangas, Philippines
tyronepaolovgarcia@iskolarngbayan.pup.edu.ph

Jowella Marie C. Layao
Polytechnic University of the Philippines
Sto. Tomas City, Batangas, Philippines
jowellamarieclayao@iskolarngbayan.pup.edu.ph

Jian Louise D. Pelayo
Polytechnic University of the Philippines
Sto. Tomas City, Batangas, Philippines
jianlouisedpelayo@iskolarngbayan.pup.edu.ph

Jhon Mack C. Robledo
Polytechnic University of the Philippines
Sto. Tomas City, Batangas, Philippines
jhonmackcrobledo@iskolarngbayan.pup.edu.ph

2025 6th International Conference on Big Data Analytics and Practices (IBDAP) | 979-8-3315-9474-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/IBDAP65587.2025.11145744

***Abstract*—This paper introduces DrugWiser, a recommended system to be implemented using machine learning to provide personalized recommendations of medicine and analyze side effects at the same time. The current study makes use of predictive models such as K-Nearest Neighbors (KNN), Decision Tree Regressor (DTR), Random Forest Regressor (RFR), Support Vector Machine (SVM), Gradient Boosting Regressor (GBR), Ada Boosting Regressor (ABR), and Light Gradient Boosting Machine (LGBM). Random Forest Regressor (RFR) performed the best with its accuracy of 99.98 percent in cross-validation and hold-out validation. To come up with a hybrid recommendation system, the model was combined with content filtering via TF-IDF and Cosine Similarity, and FuzzyWuzzy Matching when the inputs were incomplete. The final prototype was deployed with Raspberry Pi 5 that used Tkinter-based graphical user interface (GUI). Further integration with electronic health records in real time will be a part of future work to develop a better personalized medicine recommendation.**

Keywords—machine learning, medicine recommendation, medicine recommendation system, personalized medicine recommendation, random forest regressor

I. INTRODUCTION

Today's fast-paced modern technology introduced the digitization of the healthcare system to address the various issues in the healthcare sector related to medical diagnosis and medicine prescriptions. The World Health Organization has invited governments that are members to be first-movers in providing care to residents online by leveraging digital technology to enhance health and social care services [1]. The World Wide Web or Internet search engines are recognized by primary care providers as a main source of health and medical information for many patients, assisting them in evaluating the quality of medical information available online. [2]. Some surveys like the Pew Internet Survey say that over 55% of people have the Internet and rely on Web searches to get health-related information [3]. The responsibility of many healthcare professionals to acknowledge patients' decisions may change due to the addition of online health information [4]. Such generic searches do not consider the individual differences and the health suggestions are poor, in most cases.

Online buying of medicine has become very popular in the health system market [5]. Nevertheless, healthcare workers are currently overwhelmed by the sheer amount of available drugs [6], and most patients cannot pinpoint which medication is best suited to address their illness. Medicine recommendation systems should seek to solve this by giving predictions of user preferences and personal suggestions [7]. Although current developments have been embracing prescription practices, a majority of these systems are not very personalized and do not submit any side effect review according to the require preferences of the users. Also, the majority of the available research is on the web-based database and conceptual model, which has problems with practical implementation of hardware.

The study has created a recommendation system of personalized medicine which contains potential side effects and dose variants. It runs on machine learning, and therefore provides better, personalized recommendations than traditional systems can. The portable solution is based on Raspberry Pi 5 and is powered by Tkinter-based GUI, which facilitates better medication decisions users can make to boost digital healthcare development.

A. Problem Statement

Without appropriate guidance, the selection of the right medicine may sound confusing and unsafe. Having many offers related to dosage, ingredients and brands, many users cannot make informed choices [8]. Without the use of individual recommendations by symptoms or diseases, there is a risk of choosing medications that will be harmful. The current systems also do not focus on user preferences such as regard to ingredients, reviews and manufacturers as well [9], which results in insufficient instructions and possible safety concerns. Also, some of them do not have user-friendly interfaces, and it takes a long time to identify appropriate medications [10].

B. Objectives

The goal of the project, DrugWiser, is to assist the users to choose the best and safest medicine, depending on their symptoms, desired composition, rating, manufacturer, and dosage. It involves sophisticated computational techniques to produce customized suggestions such as possible side effects,

Authorized licensed use limited to: VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on February 28,2026 at 17:07:06 UTC from IEEE Xplore. Restrictions apply.

health state warnings or conflicts of drugs. It has a simple user-friendly interface to enter the symptoms comfortably and get quick and precise suggestions. Information about every medicine, such as names, side effects, and imagery, can also be found in the application to help make informed medical decisions.

C. *Significance of the Study*

This research aims to come up with a tool that can recommend the correct medicines to mitigate medication errors and enhance safer healthcare. It corresponds to SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 10 (Reduced Inequalities) given how AI and machine learning have assisted in development of inclusive and user-friendly medical support tools. The DrugWiser is aimed at the general population, particularly users who require fast and safer suggestions to common conditions. It integrates ease of use with portable devices to facilitate the correct advice on medicines [11].

II. Review of Related Works

In medicine, machine learning-based medicine recommendation systems are becoming available, however, there is little research into the use of personalized medicine recommendations. The available studies have differed in their accuracies, which led to the researchers in using their approach.

In one earlier study, drug reviews and data were collected in ICR (Ask a Patient), ICR (UCI ML Drug Review dataset), Drugs.com, and Drug Bank [6]. It implemented models of Logistic Regression, Perceptron, and LSTM with cross-validation, yet these are not too accurate. Another study involved 1,100 patient records containing 30 features and 10 classes of medicines to which it assigned Random Forest and Decision Tree classifiers [12]. In contrast to these classification-centered techniques, DrugWiser calculates customized medicine and side impact hints using regression models. It also uses NLP to identify drugs that might suit each different illness to enable better medical decisions [13].

An analysis utilizing the Drug Reviews Dataset on the UCI ML repository (Drugs.com) was able to provide 93% accuracy when utilizing Linear SVC and TF-IDF, with the lowest being Decision Tree at 78%; Word2Vec was able to achieve 88%, and Random Forest 90% [10]. In contrast to this sentiment analysis solution, DrugWiser relies on regression models and the filtering contents and fuzzy matching to recommend personalized medicines and side effects. A research based on NLP, RNN on 161,297 reviews on drugs by 102,514 users on 3,519 drugs had an accuracy of 82.6, precision of 82.8, recall of 80.6, F1-score of 81.7, and AUC of 89.3 [14]. Unlike [14], DrugWiser concentrates on regression to give specific medical information.

On Songklanagarind Hospital data, a classified collaborative filtering drug recommendation system reached 76.61% recall, 46.20% precision, and 47.50% physician agreement [15]. A different model using Kaggle Drug Review Dataset via Google Colab exhibited 80.5 accuracy on Logistic Regression and hybrid filtering [16]. In contrast to [15] and [16], the personalized medicine and side effects predictions are based on regression models using content-based filtering and fuzzy matching in DrugWiser. K-Nearest Neighbor and Decision Trees were implemented to propose drugs on the origins of the symptoms and cosine similarity [17]. Another research involved EHRs with collaborative filtering along with Decision Trees, Random Forest, and Neural Networks to classify diseases and recommend drugs [18]. The graph-based model, TriGCN, used the patient-disease-medicine data in Drugs.com [19], whereas Decision Trees and Random Forests, and clustering showed an overall performance of 88.7% - 99.3% accuracy [20]. On the contrary, DrugWiser applies regression-based prediction that contains content filtering and fuzzy matching which is less rigid and practical.

A dataset of diabetic patients consisting of more than 3000 instances was employed to develop a collaborative filtering based medicine recommendation system with inputs such as age, blood pressure, BMI and cholesterol [21]. The accuracy of disease prediction was 97.75% with SVC and Random Forest and 99% on smaller datasets using CNN and fuzzy logic [22]. DrugWiser, in turn, operates with bigger data and is aimed at providing much more accurate and reliable personalized medicine recommendation.

The reviewed studies guided the researchers to use a different dataset and aim for higher accuracy. This study involves portable devices and machine learning techniques to give a proper suggestion of medicine depending on the user input.

III. Methodology

Researchers adhered to the systematic machine learning method involving the data collection to the hardware implementation [26]. The diagram in Fig. 1 illustrates system overview that presents the major processes in the creation of the prototype based on machine learning.

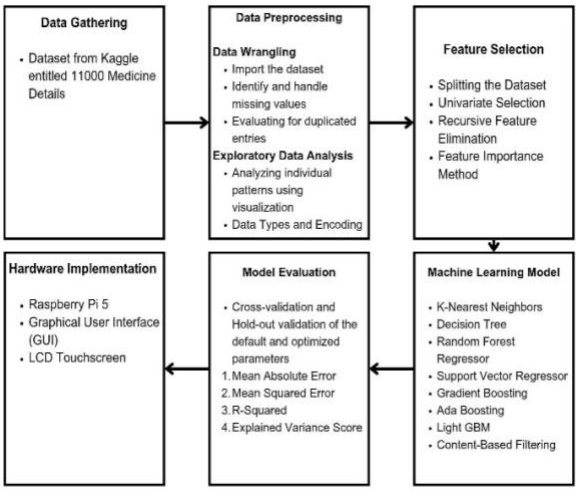

Fig. 1. System Overview

Authorized licensed use limited to: VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on February 28,2026 at 17:07:06 UTC from IEEE Xplore. Restrictions apply.

A. Dataset

The researcher utilized existing medicine datasets from Kaggle, a data science platform for available datasets online. The given dataset consists of 11,825 data and does not contain any missing values. This database includes nine features which are Medicine Name, Composition, Uses, Side Effects, Image URL, Manufacturer, Excellent Reviews, Average Reviews, and Poor Reviews [23]. This dataset is from 1mg.com a healthcare platform based in Gurugram, India.

TABLE I. COUNT AND DATA TYPE OF ALL FEATURES

Range Index: 11825 samples, 0 to 11825			
#	*Column*	*Count*	*Data Type*
0	Medicine Name	11825	object
1	Composition	11825	object
2	Uses	11825	object
3	Side_effects	11825	object
4	Image URL	11825	object
5	Manufacturer	11825	object
6	Excellent Review (%)	11825	int64
7	Average Review (%)	11825	int64
8	Poor Review (%)	11825	int64
Dtypes: int64 (3), object (6)			

B. Data Wrangling

The proponents of this research import the dataset to the Anaconda application utilizing Jupyter Notebook to distinguish if there are missing values in all categorical and numerical features. Dropping the duplicate entries in each feature reduced the numbers of the dataset, resulting in 11,741 values with no missing values. The data types of datasets are corrected respectively.

C. Exploratory Data Analysis

During this step, the researchers isolated individual feature patterns by isolating the dosage amounts in the composition creating a new feature, *Dosage Amount*. They also separate the dosage form and the name of medicine leading to the column of *Dosage Form*. Also, there is the addition of *Medicine Score*, according to users' review, to facilitate the suggestions of medicines. These modifications made the number of features to be 12 as compared to the initial nine features.

TABLE II. COUNT AND DATA TYPE OF ALL FEATURES AFTER EDA

Range Index: 11825 samples, 0 to 11741			
#	*Column*	*Count*	*Data Type*
0	Composition	11741	object
1	Uses	11741	object
2	Dosage Form	11741	object
3	Dosage Amount	11741	object
4	Manufacturer	11741	object
5	Excellent Review (%)	11741	int64
6	Average Review (%)	11741	int64
7	Poor Review (%)	11741	int64
8	Medicine Score	11741	int64
9	Image URL	11741	object
10	Side Effects	11741	object
11	Medicine Name	11741	object
Dtypes: int64 (4), object (8)			

D. Feature Selection

For this part, the dataset was splitted into two–features and targets. Different methods were used to get the top selected features including univariate selection, recursive feature elimination, and feature importance method.

E. Machine Learning Modeling

The studies contrasted a variety of models, such as Random Forest, Support Vector Machine (SVM), K-Nearest Neighbor, Decision Tree, Gradient Boosting, Light Gradient Boosting Machine (Light GBM), and Ada Boosting to perform regression. The best model in predicting medicine scores was Random Forest, which would be used, as opposed to Support Vector Regressor and Light GBM. Its flexible parameters and the fact that it does not cause overfitting makes it very useful. Cosine Similarity and TF-IDF were also applied by the researcher Content-Based Filtering [24] [25], and Fuzzywuzzy Matching to roll back in case the entry of the user lacks completeness. The hybrid approach makes a variety of approaches possible in the recommended system.

The primary software to train and develop the models involved the use of Python. To determine the effectiveness of each of the models, the dataset was divided into 80 and 20 per cent of training and testing respectively. The chosen models of machine-learning were applied to both the training and testing data. The researchers employed RandomSearch as a hyperparameter tuner, allowing exploration of a fairly larger configuration, helping produce better results on the model [27].

F. Model Evaluation

The evaluation of a regression model requires analysis through performance metrics that consist of mean absolute error and mean squared error along with r-squared and explained variance score. The system's performance and accuracy level were determined by these performance metrics, also the precision of the additional models in the

Authorized licensed use limited to: VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on February 28,2026 at 17:07:06 UTC from IEEE Xplore. Restrictions apply.

recommendation system. Once the models were already executed and compared, the model was implemented for embed code.

G. Hardware Implementation

The Raspberry Pi 5 Module was used to support the machine learning-based prototype. The system was also designed by Tkinter to provide a GUI through which a consumer could interact with its content in a friendly manner and the said GUI was loaded on Raspberry Pi and connected with 7-inch LCD touchscreen to provide accessibility. The final prototype was set up within a plywood casing and all the components sealed.

IV. Results And Discussion

A. Selected Features for Machine Learning

To prevent overfitting and improve the accuracy of the model, feature selection has been made where different methods such as univariate selection, recursive features elimination, and feature importance methods. Shown in Fig. 2 is the result of the top selected features by using univariate selection.

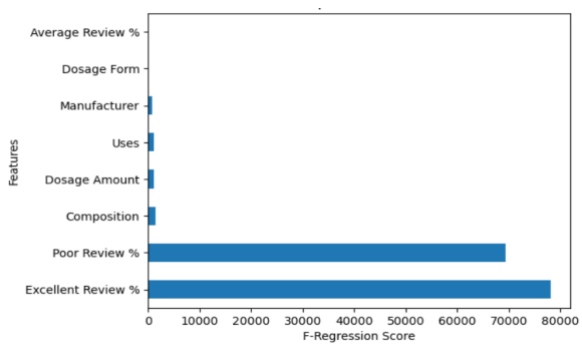

Fig. 2. Univariate Selection

For a more comprehensive analysis, shown in Fig. 3 are the top selected features by using recursive feature elimination.

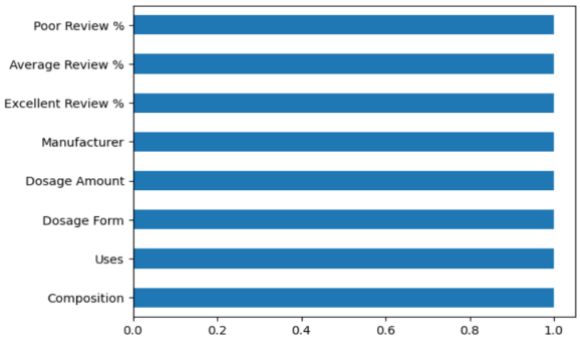

Fig. 3. Recursive Feature Elimination

This figure shows that the top features of the above-mentioned method are Composition, Uses, Dosage Form, Dosage Amount, Manufacturer, Excellent Review %, Average Review %, and Poor Review %.

Another method named the feature importance method also gave the top selected features which are shown in Fig. 4.

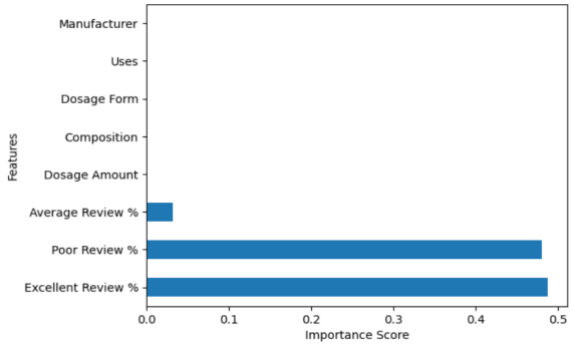

Fig. 4. Feature Importance Method

Fig. 4 shows that the Excellent Review %, Poor Review %, Average Review %, Dosage Amount, Composition, Dosage Form, Uses, and Manufacturer are the top features.

As presented in figures 2, 3, and 4, eight of the 12 features were unanimously picked as the most important. These are Composition, Uses, Dosage form, Dosage Amount, Manufacturer, Excellent Review %, Average Review %, and Poor Review %. Their frequent occurrence in various rankings emphasizes their evaluation importance, although they were a few changes in their sequence.

B. Performance of Machine Learning Models

In multiple regression frameworks, researchers applied both default and optimized parameters applying RandomSearch to K-Nearest Neighbors (KNN), Decision Tree (DTR), Random Forest (RFR), Support Vector (SVR), and Gradient Boosting (GBR), Ada Boosting (ABR), and LightGBM (LGBM). Cross-validation and hold-out validation were used during performance review of models.

The results of the separate R^2 values of default and optimized models are given in Table and Graph. The models are denoted by acronym KNN, DTR, RFR, SVR, GBR, ABR and LGBM.

Table III shows the summary of results for both cross-validation and hold-out validation of all the models in default and optimized parameters.

TABLE III. Summary of Results for R-squared Score

R-squared Score				
Models	**Cross Validation**		**Hold-out Validation**	
	Setting = Default	**Setting = Optimized**	**Setting = Default**	**Setting = Optimized**
KNN	0.9851	0.9999	0.9933	0.9941
DTR	0.9997	0.9997	0.9997	0.9993
RFR	0.9999	0.9998	0.9998	0.9997

Authorized licensed use limited to: VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on February 28,2026 at 17:07:06 UTC from IEEE Xplore. Restrictions apply.

R-squared Score				
Models	Cross Validation		Hold-out Validation	
	Setting = Default	Setting = Optimized	Setting = Default	Setting = Optimized
KNN	0.9851	0.9999	0.9933	0.9941
SVR	0.9961	0.9999	0.9968	0.9999
GBR	0.9996	0.9999	0.9995	0.9996
ABR	0.9862	0.9879	0.9874	0.9882
LGBM	0.9999	0.9999	0.9998	0.9998

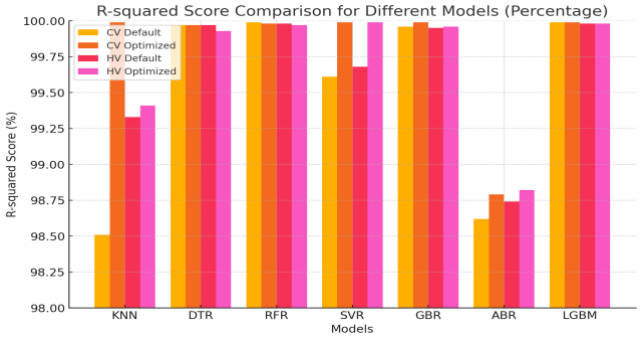

Fig. 5. Graphical Representation for R-squared Score Results

The findings demonstrated that both Random Forest and LightGBM provided the best accuracy, and it reached 99.98 and 99.99, in cross and hold-out validation, respectively. Such models as KNN, SVR, and GBR improved a bit after tuning parameters. Meanwhile, ABR had the worst accuracy at 98.7-98.8 percent. In general, the medication recommendation accuracy was high because all models worked well.

Table IV represents the results for precision, recall, f1-score, mean reciprocal rank, mean average precision, and NDCG in content-based filtering and fuzzy-wuzzy matching.

TABLE IV. SUMMARY OF RESULTS FOR CONTENT-BASED FILTERING AND FUZZYWUZZY MATCHING

Metrics	Content-Based Filtering	FuzzyWuzzy Matching
Precision	0.67	1.0
Recall	0.78	0.777
F1-score	0.71	0.866
Mean Reciprocal Rank (MRR)	1.0	1.0
NDCG	0.78	1.0
Mean Average Precision (MAP)	0.65	0.777

According to the results, content-based filtering provided quality recommendations (67 percent) and found 78 percent of all relevant medicines and precision was about equal to recall. The fuzzywuzzy matching did not make any errors in its recommended medicines, but still mistakenly missed out in some relevant ones, but still a high-balance precision and recall.

C. *Developed Hardware*

Shown in Fig. 6 is the GUI of the system.

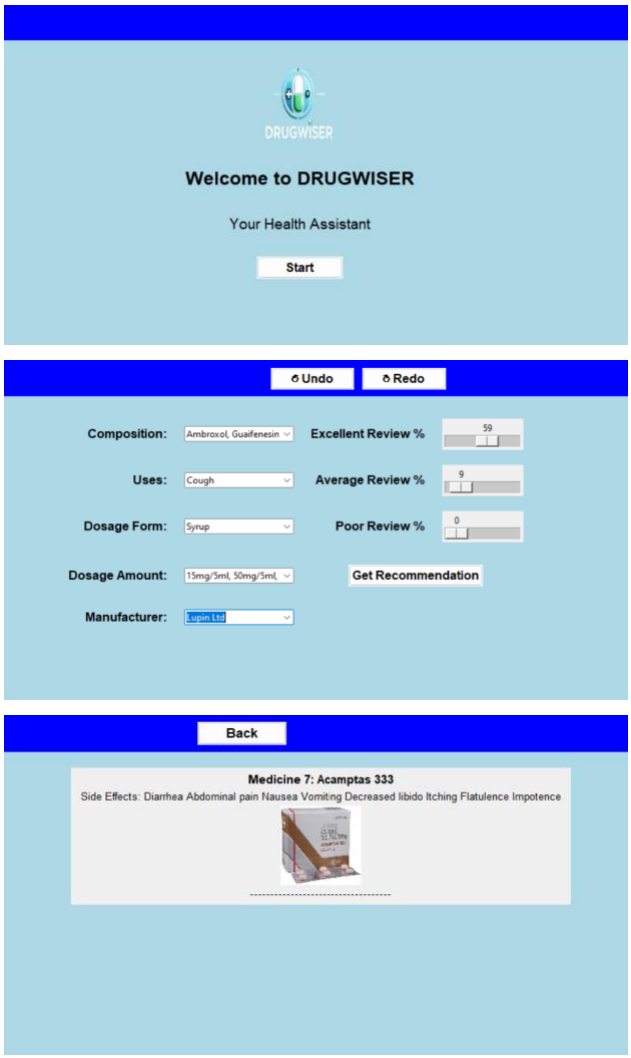

Fig. 6. Graphical User Interface of the System

Hardware implementation is shown in Fig. 7 where the Random Forest Model is embedded.

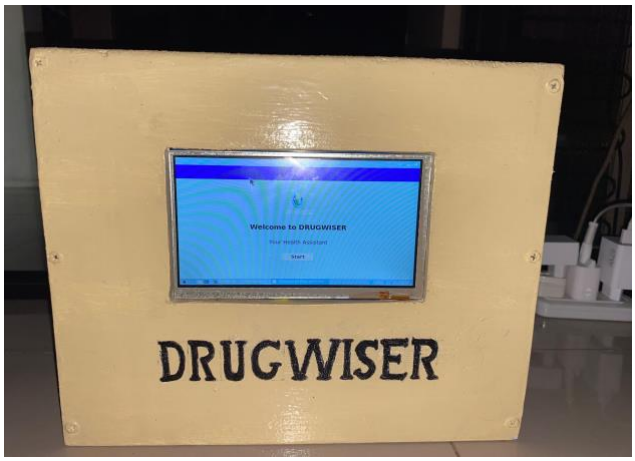

Fig. 7. Actual DrugWiser System

Authorized licensed use limited to: VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on February 28,2026 at 17:07:06 UTC from IEEE Xplore. Restrictions apply.

V. CONCLUSION

This study implemented a machine learning model which suggests medicine on the basis of the information user's input. The tested models ranked Random Forest as the best model with an accuracy of 99.98%, precision of Content-Based filtering to 67 percent and Fuzzy Wuzzy Matching to 100 percent. The system has been more accurate and hardware efficient than its predecessors.

However strong the system is, it does have its limits. It does not use medical records, test results and does not possess such features as real-time tracking, doctor consultations, and online purchases, rather making general suggestions. It also has limitations where it only has typical drugs and works with a fixed set of data that has simple data like names, pictures, and side effects.

Its potential future developments could be the implementation of electronic health records that would provide more tailored suggestions, increased rare drug coverage, the symptom monitoring capabilities offered by wearables, as well as the implementation of clinician consultation services, the multilingual interface and links to verified pharmacies.

Altogether, the researchers were able to develop a machine learning-powered system that is capable of recommending medicines correctly in regards to user preference.

VI. ACKNOWLEDGMENT

The researchers sincerely value the Polytechnic University of the Philippines providing an enabling platform that enables meaningful research and innovation. This research depended on both vital resources and appropriate scholarly works to achieve its execution goals successfully.

REFERENCES

[1] World Health Organization. (2019). Draft global strategy on digital health 2020–2024. *World Health Organization*.

[2] Diaz, J. A., Griffith, R. A., Ng, J. J., Reinert, S. E., Friedmann, P. D., & Moulton, A. W. (2002). Patients' use of the Internet for medical information. Journal of General Internal Medicine, 17(3), 180–185.

[3] Pew Internet and American Life Project. (2000, November 26). The online health care revolution: How the web helps Americans take better care of themselves. Pew Research Center.

[4] Chen, Y. Y., Li, C. M., Liang, J. C., & Tsai, C. C. (2018). Health information obtained from the Internet and changes in medical decision making: Questionnaire development and cross-sectional survey. Journal of Medical Internet Research, 20(2), e47.

[5] Almomani, H., Raza, A., Patel, N., & Donyai, P. (2023). Reasons that lead people to buy prescription medicines on the internet: A systematic review. Frontiers in Pharmacology, 14, 1239507.

[6] Keikhosrokiani, P., Balasubramaniam, K., & Isomursu, M. (2024). Drug recommendation system for healthcare professionals' decision-making using opinion mining and machine learning. In M. Särestöniemi et al. (Eds.), Digital health and wireless solutions (Vol. 2084, pp. 233-246). Springer.

[7] Bhat, S., & Aishwarya, K. (2013). Item-based hybrid recommender system for newly marketed pharmaceutical drugs. In 2013 International Conference on Advances in Computing, Communications and Informatics (ICACCI) (pp. 2107–2111). IEEE.

[8] Marin, I., Goga, N., & Stanciu, R. C. (2019). Web application for self-diagnosis and drug recommendation based on user symptoms. Journal of Advances in Technology and Engineering Research, 5(2), 62–71.

[9] Shambour, Q., Al-Zyoud, M., Abu-Shareha, A., & Abualhaj, M. (2023). Medicine recommender system based on semantic and multi-criteria filtering. Interdisciplinary Journal of Information, Knowledge, and Management, 18, 435–457.

[10] Garg, S. (2021). Drug recommendation system based on sentiment analysis of drug reviews using machine learning. In 2021 11th International Conference on Cloud Computing, Data Science and Engineering (Confluence). IEEE.

[11] A. U. Observatory. (2020). Using the AUO to help your community meet the United Nations sustainable development goals. Sustainable Development Goals.

[12] Balakrishna, N. C., & Srinivasulu, N. S. (2024). Drug suggestion mechanism in medical emergencies using machine learning. International Journal of Science and Research Archive, 13(1), 901–906.

[13] Swain, K. P., Mohapatra, S., Ravi, V., Nayak, S. R., Alahmadi, T. J., Singh, P., & Diwakar, M. (2024). Leveraging machine learning and patient reviews for developing a drug recommendation system to reduce medical errors. The Open Bioinformatics Journal, 17(1).

[14] Begum, S. G., & Sree, P. K. (2023). Drug recommendations using a "reviews and sentiment analysis" by a recurrent neural network. Indonesian Journal of Multidisciplinary Science, 2(9), 3085–3094.

[15] Sae-Ang, A., Chairat, S., Tansuebchueasai, N., Fumaneeshoat, O., Ingviya, T., & Chaichulee, S. (2022). Drug recommendation from diagnosis codes: Classification vs. collaborative filtering approaches. International Journal of Environmental Research and Public Health, 20(1), 309.

[16] Mohapatra, M., Nayak, M., & Mahapatra, S. (2022). A machine learning-based drug recommendation system for health care. Graduate Research in Engineering and Technology, 5–10.

[17] Saranya, K., Jayashree, R., & Ranjitha, A. (2023). Recommendation of drugs and its substitute medicines also purchasing using cosine similarity vector. International Journal of Scientific Research in Science, Engineering and Technology, 28–33.

[18] S, A. H., M, A., G, K. N., BR, T., & Kashyap, V. M. (2024). Personalized drug recommender system to aid medical professionals. IARJSET, 11(6).

[19] Zhou, H., Liao, S., & Guo, F. (2024). TRIGCN: Graph convolution network based on tripartite graph for personalized medicine recommendation system. Systems, 12(10), 398.

[20] Lavanya, P., Reddy, Y. S. P., Reddy, R. R., & Reddy, S. R. (2024). Personalized medicine recommendation system using machine learning. International Journal of Engineering Innovations and Management Strategies, 1(2).

[21] Kakulapati, V., Balaram, V. V. S. S. B., & Reddy, S. M. (2018). Collaborative filtering recommendation systems for personalized medicines. International Journal of Engineering Research in Computer Science and Engineering (IJERCSE), 5(4).

[22] Hassan, B. M., & Elagamy, S. M. (2025). Personalized medical recommendation system with machine learning. Neural Computing and Applications.

[23] Singh, N. (2024). 11000 medicine dataset details. Kaggle.

[24] Luna, R. G. D., Cuizon, A. V. O., Umali, J. P. D., Latayan, E. M., Onte, M. J. M., & Ortega, H. M. (2024). Pawfect Match: A user-interactive dog breed recommendation powered by machine learning. In 2024 7th International Conference on Informatics and Computational Sciences (ICICoS) (pp. 97–102). IEEE.

[25] De Luna, R. G., Batingan, P. C., Berbon, B. B., Kish, M. J., Pecayo, K. M., & Villegas, R. J. L. (2024). SPECTRA Phones: Systematic personalized evaluation for comprehensive technological recommendations and advancements in phones. In 2024 7th International Conference on Informatics and Computational Sciences (ICICoS) (pp. 115–120). IEEE.

[26] De Luna, R. G., Biscocho, K. B., Rosario, C. A. A. D., Elipse, M. M. C. O., Pecho, P. O., & Silvestre, S. N. O. (2024). Soil Analysis-Guided Expert: A machine learning device for optimized fertilizer selection. In 2024 7th International Conference on Informatics and Computational Sciences (ICICoS) (pp. 109–114). IEEE.

[27] Ali, Y. A., Awwad, E. M., Al-Razgan, M., & Maarouf, A. (2023). Hyperparameter Search for Machine Learning Algorithms for Optimizing the Computational Complexity. Processes, 11(2), 349

Authorized licensed use limited to: VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on February 28,2026 at 17:07:06 UTC from IEEE Xplore. Restrictions apply.